Matrices & Vectors

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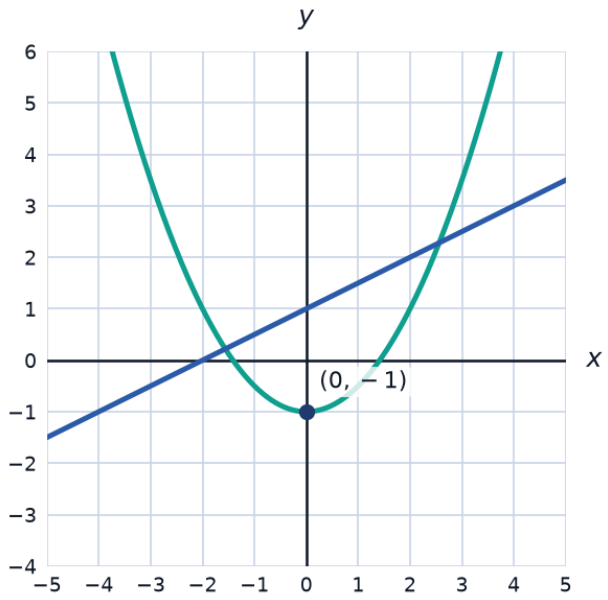
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and B be the matrix with rows $\langle 2, -1\rangle$ and $\langle 0, 5\rangle$. Find $A + B$ and $2A$.

2 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 1, 3\rangle$, $\langle -2, 4\rangle$) and B (rows $\langle 2, -1\rangle$, $\langle 0, 5\rangle$) from above, find AB.



3 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and let $\mathbf{v} = \langle 3, -1\rangle$. Find $A\mathbf{v}$.

4 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and state whether A is invertible.

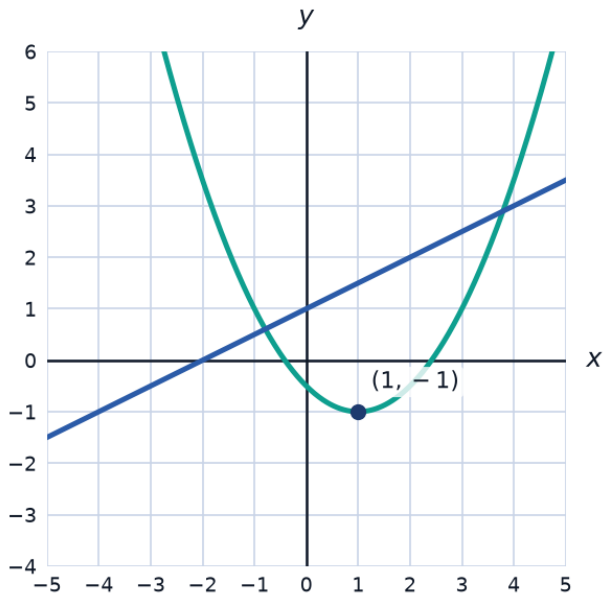
5 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$.

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6 [Foundational] • Target: Solving systems with the inverse matrix

Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).



7 [Foundational] • Target: Vector magnitude and dot product

For $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{w} = \langle -3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.

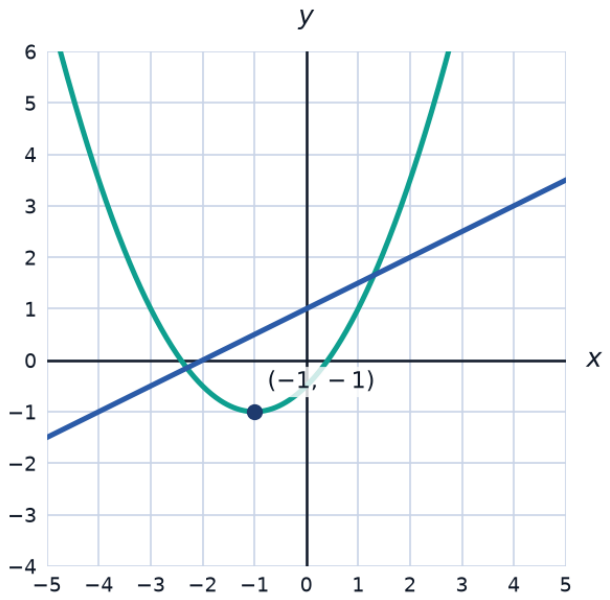
9 [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.

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10 [Intermediate] • Target: 3D Vectors & Cross Products

Find the angle θ between $\mathbf{u} = \langle -6, -7 \rangle$ and $\mathbf{v} = \langle 2, 6 \rangle$, to the nearest 0.1°.



11 [Intermediate] • Target: Addition and scalar multiplication

Find the angle θ between $\mathbf{u} = \langle 6, 8 \rangle$ and $\mathbf{v} = \langle 7, -4 \rangle$, to the nearest 0.1°.

12 [Intermediate] • Target: Matrix multiplication

Find the angle θ between $\mathbf{u} = \langle 6, -7 \rangle$ and $\mathbf{v} = \langle -3, 0 \rangle$, to the nearest 0.1°.

13 [Intermediate] • Target: Determinant and inverse

Find the angle θ between $\mathbf{u} = \langle -5, 5 \rangle$ and $\mathbf{v} = \langle -3, 8 \rangle$, to the nearest 0.1°.

14 [Intermediate] • Target: Solving systems with the inverse matrix

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

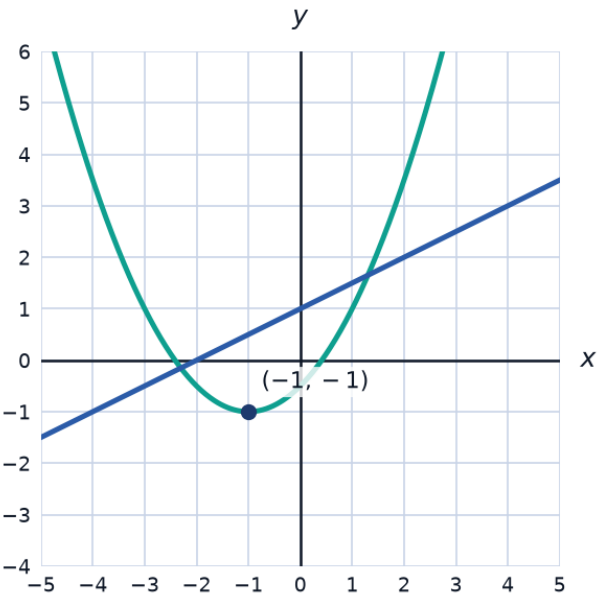
15 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 3.01 Functions & Graphs

Find the product of the roots of $x^2 + 14x + 113 = 0$.

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16 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Introductory Calculus

Find the product of the roots of $x^2 - 18x + 106 = 0$.



17 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 6.01 Probability

Let $z = 3 - 4i$. What is the real part of z?

18 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

Find A^{-1} for $A = [-3, -6; -5, -6]$.

19 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

Find A^{-1} for $A = [2, -5; -1, -2]$.

20 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Does $y = 6x^2 + 9$ solve $\frac{dy}{dx} = 12x$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Addition and scalar multiplication

Let A be the matrix with rows $\langle -8, 4\rangle$ and $\langle 7, 7\rangle$, and B be the matrix with rows $\langle -2, 4\rangle$ and $\langle -7, -3\rangle$. Find $A + B$.

a. $\langle -8, 8\rangle, \langle 0, 4\rangle$

b. $\langle -10, 8\rangle, \langle 0, 2\rangle$

c. $\langle -10, 8\rangle, \langle 0, 4\rangle$

d. $\langle 0, 4\rangle, \langle -10, 8\rangle$

22 [Foundational] • Target: Matrix multiplication

Let A be the matrix with rows $\langle -8, -2\rangle$ and $\langle 8, 8\rangle$, and B be the matrix with rows $\langle -4, -8\rangle$ and $\langle -5, 5\rangle$. Find $A + B$.

a. $\langle 3, 13\rangle, \langle -12, -10\rangle$

b. $\langle -10, -10\rangle, \langle 3, 13\rangle$

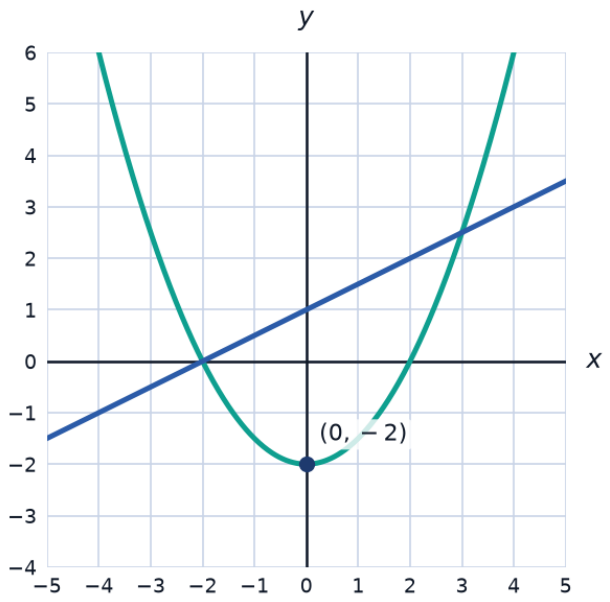
c. $\langle -12, -10\rangle, \langle 3, 13\rangle$

d. $\langle -12, -10\rangle, \langle 3, 11\rangle$

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23 [Foundational] • Target: Determinant and inverse

Let A be the matrix with rows $\langle 1, -3 \rangle$ and $\langle 7, 6 \rangle$, and B be the matrix with rows $\langle 6, 2 \rangle$ and $\langle -5, 1 \rangle$. Find $A + B$.



a. $\langle 7, -1 \rangle, \langle 2, 5 \rangle$

b. $\langle 2, 7 \rangle, \langle 7, -1 \rangle$

c. $\langle 7, -1 \rangle, \langle 2, 7 \rangle$

d. $\langle 9, -1 \rangle, \langle 2, 7 \rangle$

24 [Foundational] • Target: Solving systems with the inverse matrix

Let A be the matrix with rows $\langle 6, -2 \rangle$ and $\langle -8, -6 \rangle$, and B be the matrix with rows $\langle -5, -4 \rangle$ and $\langle 5, -6 \rangle$. Find $A + B$.

a. $\langle 1, -6 \rangle, \langle -3, -12 \rangle$

b. $\langle 3, -6 \rangle, \langle -3, -12 \rangle$

c. $\langle -3, -12 \rangle, \langle 1, -6 \rangle$

d. $\langle 1, -6 \rangle, \langle -3, -14 \rangle$

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25 [Foundational] • Target: Vector magnitude and dot product

Let A be the matrix with rows $\langle -4, 3\rangle$ and $\langle -7, 1\rangle$, and B be the matrix with rows $\langle -2, 9\rangle$ and $\langle -5, 1\rangle$. Find $A + B$.

a. $\langle -4, 12\rangle, \langle -12, 2\rangle$

b. $\langle -6, 12\rangle, \langle -12, 0\rangle$

c. $\langle -12, 2\rangle, \langle -6, 12\rangle$

d. $\langle -6, 12\rangle, \langle -12, 2\rangle$

26 [Foundational] • Target: Angle between vectors

Let A be the matrix with rows $\langle 8, -7\rangle$ and $\langle 9, -7\rangle$, and B be the matrix with rows $\langle 7, 9\rangle$ and $\langle 1, 9\rangle$. Find $A + B$.

a. $\langle 17, 2\rangle, \langle 10, 2\rangle$

b. $\langle 10, 2\rangle, \langle 15, 2\rangle$

c. $\langle 15, 2\rangle, \langle 10, 2\rangle$

d. $\langle 15, 2\rangle, \langle 10, 0\rangle$

27 [Foundational] • Target: Matrix Inversion & Systems

Let A be the matrix with rows $\langle -6, 5\rangle$ and $\langle -9, 1\rangle$, and B be the matrix with rows $\langle 8, 3\rangle$ and $\langle 7, 6\rangle$. Find $A + B$.

a. $\langle 4, 8\rangle, \langle -2, 7\rangle$

b. $\langle 2, 8\rangle, \langle -2, 5\rangle$

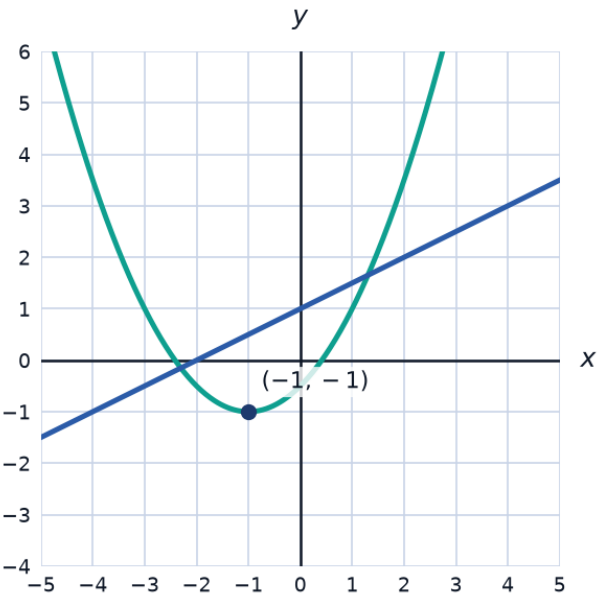
c. $\langle 2, 8\rangle, \langle -2, 7\rangle$

d. $\langle -2, 7\rangle, \langle 2, 8\rangle$

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28 [Foundational] • Target: Determinants & Properties

Let A have rows $\langle -9, 8\rangle$ and $\langle -9, 6\rangle$, and let $\mathbf{v} = \langle 6, 3\rangle$. Find $A\mathbf{v}$.



a. $\langle -36, -30\rangle$

b. $\langle -28, -36\rangle$

c. $\langle -30, -36\rangle$

d. $\langle -30, -38\rangle$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Dot Product & Projections

Let A have rows $\langle 9, -1\rangle$ and $\langle 4, -1\rangle$, and let $\mathbf{v} = \langle 3, -5\rangle$. Find $A\mathbf{v}$.

a. $\langle 17, 32\rangle$

b. $\langle 32, 15\rangle$

c. $\langle 34, 17\rangle$

d. $\langle 32, 17\rangle$

30 [Intermediate] • Target: 3D Vectors & Cross Products

Let A have rows $\langle -4, 7\rangle$ and $\langle 6, 4\rangle$, and let $\mathbf{v} = \langle -8, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 69, -28\rangle$

b. $\langle 67, -28\rangle$

c. $\langle -28, 67\rangle$

d. $\langle 67, -30\rangle$

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31 [Intermediate] • Target: Addition and scalar multiplication

Let A have rows $\langle -2, 2\rangle$ and $\langle 8, -1\rangle$, and let $\mathbf{v} = \langle -9, -1\rangle$. Find $A\mathbf{v}$.

a. $\langle 18, -71\rangle$

b. $\langle -71, 16\rangle$

c. $\langle 16, -73\rangle$

d. $\langle 16, -71\rangle$

32 [Intermediate] • Target: Matrix multiplication

Let A have rows $\langle 4, -6\rangle$ and $\langle 8, -7\rangle$, and let $\mathbf{v} = \langle 6, -9\rangle$. Find $A\mathbf{v}$.

a. $\langle 111, 78\rangle$

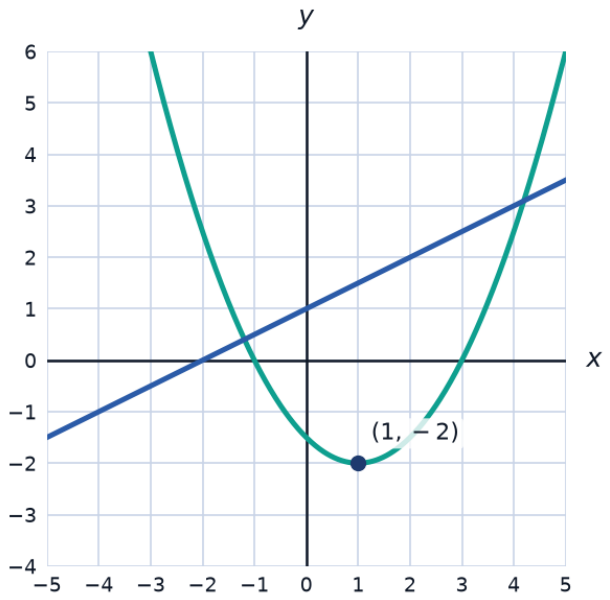
b. $\langle 78, 111\rangle$

c. $\langle 80, 111\rangle$

d. $\langle 78, 109\rangle$

33 [Intermediate] • Target: Determinant and inverse

Let A have rows $\langle -3, 4\rangle$ and $\langle -1, -6\rangle$, and let $\mathbf{v} = \langle 1, -4\rangle$. Find $A\mathbf{v}$.



a. $\langle -19, 21\rangle$

b. $\langle -17, 23\rangle$

c. $\langle -19, 23\rangle$

d. $\langle 23, -19\rangle$

34 [Intermediate] • Target: Solving systems with the inverse matrix

Let A have rows $\langle 2, -6\rangle$ and $\langle 3, 3\rangle$, and let $\mathbf{v} = \langle 3, 8\rangle$. Find $A\mathbf{v}$.

a. $\langle -42, 31\rangle$

b. $\langle -42, 33\rangle$

c. $\langle 33, -42\rangle$

d. $\langle -40, 33\rangle$

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35 [Intermediate] • Target: Vector magnitude and dot product

Find $\det A$ for the matrix A with rows $\langle -5, -3\rangle$ and $\langle 6, 4\rangle$.

a. -2

b. 2

c. -4

d. 0

36 [Intermediate] • Target: Angle between vectors

Find $\det A$ for the matrix A with rows $\langle -3, -3\rangle$ and $\langle -3, 5\rangle$.

a. -26

b. 24

c. -22

d. -24

37 [Intermediate] • Target: Matrix Inversion & Systems

For $\mathbf{u} = \langle -20, -21\rangle$ and $\mathbf{w} = \langle -8, 7\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $15, 29$

b. $13, 31$

c. $29, 13$

d. $13, 29$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

For $\mathbf{u} = \langle -12, 35\rangle$ and $\mathbf{w} = \langle -7, 3\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $191, 37$

b. $189, 37$

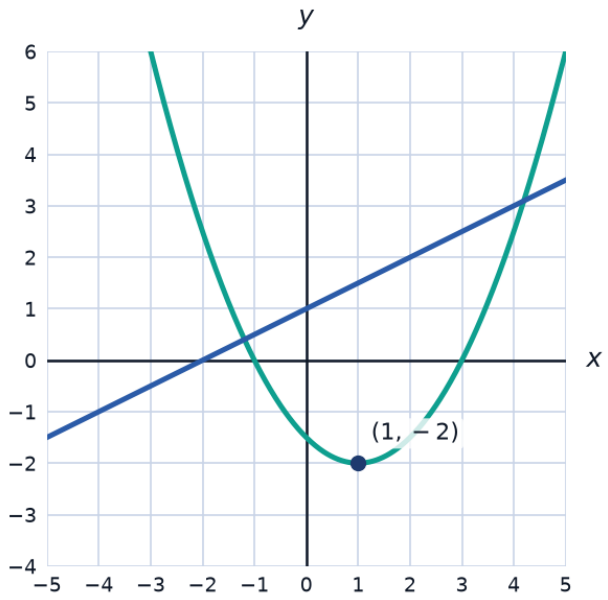
c. $37, 189$

d. $189, 39$

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39 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

Find the angle between $\langle -5, 5\rangle$ and $\langle 5, 2\rangle$, rounded to the nearest tenth of a degree.



a. 123.2°

b. 113.2°

c. 108.2°

d. 118.2°

40 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Find the angle between $\langle 4, 6\rangle$ and $\langle -6, 1\rangle$, rounded to the nearest tenth of a degree.

a. 124.2°

b. 119.2°

c. 114.2°

d. 109.2°

41 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 5.01 Introductory Calculus

What is the inverse of $A = [[4, 7], [1, 2]]$?

a. $[[2, -7], [-1, 4]]$

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

42 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 6.01 Probability

If $AX = B$, where A is invertible, which expression gives the solution column vector X?

a. $A^{-1}B$

b. BA^{-1}

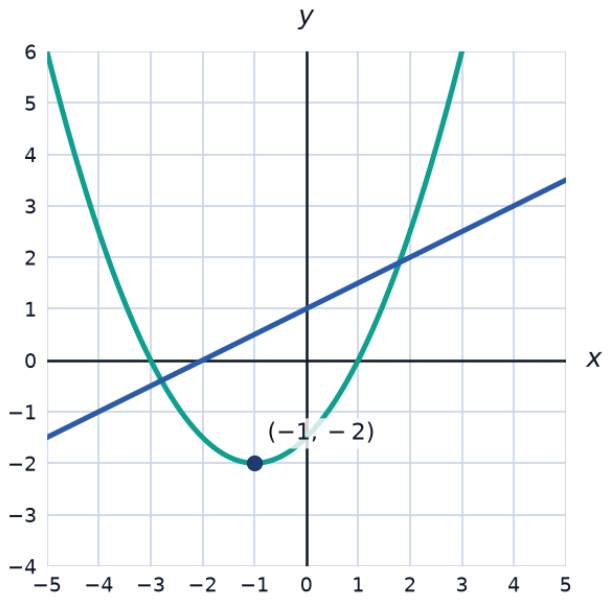
c. B/A

d. AB^{-1}

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43 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

What happens to the determinant of a matrix if two of its rows are interchanged?



a. Its sign is reversed

b. It doubles

c. It remains unchanged

d. It becomes zero

44 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

What is the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$?

a. $\langle 3, 0 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 0, 4 \rangle$

d. $\langle 1, 0 \rangle$

45 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Compute the cross product $\langle 1, 0, 0 \rangle \times \langle 0, 1, 0 \rangle$.

a. $\langle 0, 0, 1 \rangle$

b. $\langle 0, 0, -1 \rangle$

c. $\langle 1, 1, 0 \rangle$

d. $\mathbf{0}$

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